

02 Descriptives

The ‘Immigration Issue’ Misnomer in Human Capital Research: Intersectional Inequalities in Adult Education and Literacy in Upper- and Middle-Income Countries

Author Omitted for Blind Peer Review

Data prepared for this stage in 01_Data_Setup.Rmd. Except for Table A1 (new data imported)
The first cycle of PIAAC is included to demonstrate change in literacy across countries.
Many functions are stored in the /Sub folder and called in this workflow.

Code knitted as HTML, due to HTML figures/tables, therefore, color-coding not reproduced (all red) unlike
subsequent workflows.

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1 Setup

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

packages <- c('tidyverse',
              'readr',
              'knitr',
              'here',
              'ggplot2',
              'ggrepel',
              'ragg',
              'survey',
              'kableExtra',
              'scales',
              'wbstats')

pacman::p_load(packages, character.only = T)
```

2 Data

```
df <- read_rds(here("Data", "df.RDS"))
df2 <- read_rds(here("Data", "df2.RDS"))

# basic or lower literacy, drop Denmark because FORLANG is missing
df2 <- df2 %>%
  subset(iso3c != "DNK") %>%
  mutate(reading_low = ifelse(literacy_level %in% c(0, 1), 1, 0),
         ALL = 1)

df <- df %>%
  mutate(reading_low = ifelse(literacy_level %in% c(0, 1), 1, 0),
         ALL = 1)

europe <- c("AUT", "BEL", "CHE", "CZE", "DEU", "DNK", "ESP", "EST", "FIN", "FRA", "GBR", "H
RV", "HUN", "IRL", "ITA", "LTU", "LVA", "NOR", "POL", "PRT", "SVK", "SWE")

rich <- c("AUT", "BEL", "CHE", "DEU", "DNK", "ESP", "FIN", "FRA", "GBR", "IRL", "ITA", "NOR
", "PRT", "SWE", "JPN", "CAN", "USA", "AUS", "NZL")

dfeurope <- subset(df, iso3c %in% europe)
df2europe <- subset(df2, iso3c %in% europe)

dfrich <- subset(df, iso3c %in% rich)
df2rich <- subset(df2, iso3c %in% rich)

dfde <- subset(df, iso3c == "DEU")
df2de <- subset(df2, iso3c == "DEU")

# for repeated measures means
df1_rep <- svrepdesign(
  weights = ~SPFWT0,
  repweights = df[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = df)

df2_rep <- svrepdesign(
  weights = ~SPFWT0,
```

```

repweights = df2[, paste0("SPFWT", 1:80)],
type = "Fay", rho = 0.5, mse = T, data = df2)

df1_rep_eur <- svrepdesign(
  weights = ~SPFWT0,
  repweights = dfeurope[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = dfeurope)

df2_rep_eur <- svrepdesign(
  weights = ~SPFWT0,
  repweights = df2europe[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = df2europe)

df1_rep_rich <- svrepdesign(
  weights = ~SPFWT0,
  repweights = dfrich[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = dfrich)

df2_rep_rich <- svrepdesign(
  weights = ~SPFWT0,
  repweights = df2rich[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = df2rich)

df1_rep_de <- svrepdesign(
  weights = ~SPFWT0,
  repweights = dfde[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = dfde)

df2_rep_de <- svrepdesign(
  weights = ~SPFWT0,
  repweights = df2de[, paste0("SPFWT", 1:80)],
  type = "Fay", rho = 0.5, mse = T, data = df2de)

pvlit <- paste0("PVLIT", 1:10)
pvnum <- paste0("PVNUM", 1:10)

# helper function to identify rep means/SD for sub-groups
source(here("Sub", "IRT_helper_groups_se_pop.R"))

```

3 Fig A1. Immigrant Literacy 2012 and 2022

3.1 Get rates for each country

Run the routine `country_loop.R` once and then simply load the results thereafter as the saved object `plot_df.RData`

```

# Basic version

source(here("Sub", "IRT_helper_groups.R"))

# source(here("Sub", "country_loop.R"))

load(here("Data", "plot_df.RData"))

```

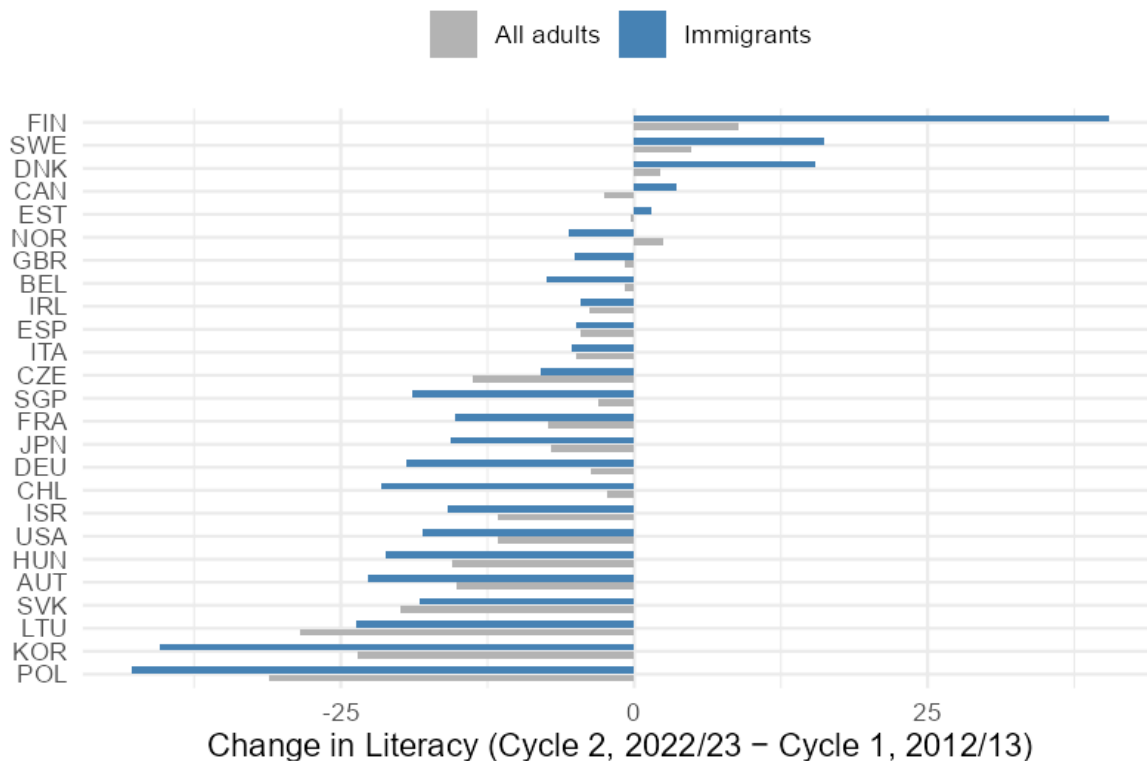
3.2 Build

```
agg_png(here("Results", "Fig1A.png"), res = 144, height = 600, width = 800)

ggplot(plot_df, aes(x = iso3c, y = change_lit, fill = group)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  coord_flip() +
  labs(
    x = NULL,
    y = "Change in Literacy (Cycle 2, 2022/23 - Cycle 1, 2012/13)",
    title = "Change in Mean Literacy between PIAAC Cycles",
    caption = "NOTE: estimated with R survey package using repeated measures and population weights",
    fill = NULL
  ) +
  scale_fill_manual(values = c("Immigrants" = "steelblue", "All adults" = "gray70")) +
  theme_minimal() +
  theme(legend.position = "top",
        plot.caption = element_text(hjust = 0, size = 6, face = "italic"))

dev.off()
## png
## 2
include_graphics(here("Results", "Fig1A.png"))
```

Change in Mean Literacy between PIAAC Cycles



NOTE: estimated with R survey package using repeated measures and population weights

4 Tbl S1. Immigration Increases 1990-2024

UN. 2024. "Total Number of Emigrants: People That Were Born in a Country and Now Live in Another Country" [CSV]. Our World In Data: United Nations Department of Economic and Social Affairs.

<https://ourworldindata.org/migration>.

UN. 2024. "Share of The Population That Were Born in Another Country" [CSV]. Our World In Data: United Nations Department of Economic and Social Affairs. <https://ourworldindata.org/grapher/migrant-stock-share>

Files downloaded 2025/12/05

```
df_un_mig <- read_csv(here("Data", "total-number-of-emigrants.filtered", "total-number-of-emigrants.csv"))

df_un_fb <- read_csv(here("Data", "migrant-stock-share.filtered", "migrant-stock-share.csv")) %>%
  subset(Year %in% c(1990, 2024))

continents <- c("Africa", "Asia", "Europe", "North America", "Oceania", "South America")

df_emig_out <- df_un_mig %>%
  filter(Entity %in% continents, Year %in% c(1990, 2024)) %>%
  select(Entity, Year, emigrants = `Total number of emigrants`) %>%
  pivot_wider(names_from = Year, values_from = emigrants) %>%
  transmute(
    Entity,
    emigrants_2024 = `2024`,
    pct_change_1990_2024 = (`2024` - `1990`) / `1990` * 100
  ) %>%
  bind_rows(
    df_un_mig %>%
      filter(Entity %in% continents, Year %in% c(1990, 2024)) %>%
      group_by(Year) %>%
      summarise(emigrants = sum(`Total number of emigrants`, na.rm = TRUE), .groups="drop")
  ) %>%
  pivot_wider(names_from = Year, values_from = emigrants) %>%
  transmute(
    Entity = "World",
    emigrants_2024 = `2024`,
    pct_change_1990_2024 = (`2024` - `1990`) / `1990` * 100
  )

)

source(here("Sub", "pop_2024.R"))

# this does not include Oceania, quick Google info 46 mil in 2024

df_pop_2024[7,1] <- "Oceania"
df_pop_2024[7,2] <- 46000000

df_2A <- df_emig_out %>%
  left_join(df_pop_2024, by = "Entity")

df_2A <- df_2A %>%
  mutate(pct_emig_2024 = round(100*(emigrants_2024/population_2024),1))

df_2A <- df_2A %>%
  mutate(pct_fb_2024 = case_when(Entity == "Africa" ~ df_un_fb$`Share of the population that was born in another country`[df_un_fb$Entity == "Africa" & df_un_fb$Year == 2024],
    Entity == "Asia" ~ df_un_fb$`Share of the population that was born in another country`[df_un_fb$Entity == "Asia" & df_un_fb$Year == 2024],
    Entity == "Europe" ~ df_un_fb$`Share of the population that was born in another country`[df_un_fb$Entity == "Western Europe (UN)" & df_un_fb$Year == 2024],
    Entity == "North America" ~ df_un_fb$`Share of the population that was born in another country`[df_un_fb$Entity == "North America" & df_un_fb$Year == 2024],
```

```

        Entity == "Oceania" ~ df_un_fb$`Share of the population th
at was born in another country`[df_un_fb$Entity == "Oceania" & df_un_fb$Year == 2024],
        Entity == "South America" ~ df_un_fb$`Share of the populat
ion that was born in another country`[df_un_fb$Entity == "South America" & df_un_fb$Year ==
2024],
        Entity == "World" ~ df_un_fb$`Share of the population that
was born in another country`[df_un_fb$Entity == "World" & df_un_fb$Year == 2024]),
    pct_fb_2024 = round(pct_fb_2024, 1),
    pct_fb_1990 = case_when(Entity == "Africa" ~ df_un_fb$`Share of the population tha
t was born in another country`[df_un_fb$Entity == "Africa" & df_un_fb$Year == 1990],
        Entity == "Asia" ~ df_un_fb$`Share of the population that
was born in another country`[df_un_fb$Entity == "Asia" & df_un_fb$Year == 1990],
        Entity == "Europe" ~ df_un_fb$`Share of the population tha
t was born in another country`[df_un_fb$Entity == "Western Europe (UN)" & df_un_fb$Year ==
1990],
        Entity == "North America" ~ df_un_fb$`Share of the populat
ion that was born in another country`[df_un_fb$Entity == "North America" & df_un_fb$Year ==
1990],
        Entity == "Oceania" ~ df_un_fb$`Share of the population th
at was born in another country`[df_un_fb$Entity == "Oceania" & df_un_fb$Year == 1990],
        Entity == "South America" ~ df_un_fb$`Share of the populat
ion that was born in another country`[df_un_fb$Entity == "South America" & df_un_fb$Year ==
1990],
        Entity == "World" ~ df_un_fb$`Share of the population that
was born in another country`[df_un_fb$Entity == "World" & df_un_fb$Year == 1990]),
    pct_fb_1990 = round(pct_fb_1990, 1),
  )

write.csv(df_2A, here("Results", "TS2.csv"), row.names = FALSE)

kable(df_2A)
rm(df_2A)

rm(df_un_mig, df_emig_out, df_pop_2024, df_un_fb)

```

Entity	emigrants_2024	pct_change_1990_2024	population_2024	pct_emig_2024	pct_fb_2024	pct_fb_1990
Africa	45750673	132.49810	1291044964	3.5	1.9	2.5
Asia	120717155	122.91332	4042291127	3.0	1.9	1.5
Europe	61061575	24.86158	928263208	6.6	18.4	9.6
North America	31580108	139.35463	381464223	8.3	10.8	7.1
Oceania	1602503	95.48417	46000000	3.5	21.5	17.8
South America	21243834	418.78533	662186388	3.2	2.9	1.5
World	281955848	100.19000	7305249910	3.9	3.7	2.9

5 Table A1. Descriptives

5.1 All

```

table_1_all <- df2 %>%
  subset(ed_currently != 1) %>%
  mutate(
    ed_primary = ifelse(ed3 == 1, 1, 0),
    ed_secondary = ifelse(ed3 == 2, 1, 0),
    ed_tertiary = ifelse(ed3 == 3, 1, 0),
    missing = case_when(is.na(skill_reading) | is.na(NFE12) | is.na(FORBORN) | is.na(children) | is.na(FORLANG) | is.na(ed3) | is.na(ed_currently) ~ 1,
                        .default = NA)
  ) %>%
  select(-ed3) %>%
  select(iso3c, skill_reading, NFE12, female, FORBORN, children, FORLANG, ed_primary, ed_secondary, ed_tertiary, missing, aweight)

# drop countries that do not have all variables
valid_countries <- table_1_all %>%
  group_by(iso3c) %>%
  summarise(across(everything(), ~ sum(!is.na(.)))) %>%
  filter(across(-iso3c, ~ . > 0)) %>%
  pull(iso3c)

table_1_all <- table_1_all %>%
  filter(iso3c %in% valid_countries)

table_louta <- table_1_all %>%
  select(-iso3c) %>%
  summarise(across(
    everything(),
    list(
      mean = ~ mean(., na.rm = TRUE),
      wmean = ~ weighted.mean(., aweight, na.rm = TRUE),
      sd = ~ sd(., na.rm = TRUE),
      min = ~ min(., na.rm = TRUE),
      max = ~ max(., na.rm = TRUE),
      n = ~ sum(!is.na(.))
    ),
    .names = "{.col}__{.fn}"
  )) %>%
  pivot_longer(
    everything(),
    names_to = c("variable", ".value"),
    names_sep = "__"
  ) %>%
  mutate(variable = case_when(
    variable == "skill_reading" ~ "Literacy Skills",
    variable == "NFE12" ~ "ALE Participation",
    variable == "FORBORN" ~ "Foreign-Born",
    variable == "children" ~ "Has Children",
    variable == "FORLANG" ~ "Non-Native Speaker",
    variable == "ed_primary" ~ "Primary Education",
    variable == "ed_secondary" ~ "Secondary Education",
    variable == "ed_tertiary" ~ "Tertiary Education",
    variable == "missing" ~ "Listwise Missing Cases",
    TRUE ~ variable
  ))

# Save as HTML
table_louta %>%
  select(variable, mean, wmean, sd, min, max, n) %>%
  kable("html", digits = 2, caption = "Descriptive Statistics with Weighted Means") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  save_kable(here("Results", "table_S1.html"))

```

```
table_louta %>% kable("html", digits = 2, caption = "Descriptive Statistics") %>%  
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))  
write.csv(  
  table_louta,  
  file = here("Results", "TA1.csv"),  
  row.names = FALSE  
)
```


Descriptive Statistics

variable	mean	wmean	sd	min	max	n
Literacy Skills	256.39	260.88	55.40	16.76	449.50	126621
ALE Participation	0.41	0.46	0.49	0.00	1.00	124856
female	0.54	0.50	0.50	0.00	1.00	126432
Foreign-Born	0.13	0.25	0.34	0.00	1.00	123375
Has Children	0.67	0.64	0.47	0.00	1.00	126621
Non-Native Speaker	0.11	0.19	0.31	0.00	1.00	123375
Primary Education	0.13	0.14	0.33	0.00	1.00	124323
Secondary Education	0.44	0.40	0.50	0.00	1.00	124323
Tertiary Education	0.44	0.45	0.50	0.00	1.00	124323
Listwise Missing Cases	1.00	1.00	0.00	1.00	1.00	3805
aweight	24.50	493.24	107.16	1.00	3988.08	126621

Drop cases of those still in formal education 20478 and do listwise deletion leaving 122816 cases for analysis. This is a 3.01 percent reduction.

5.2 Women

```
table_1 <- df2 %>%
  subset(female == 1 & !is.na(female) & ed_currently != 1) %>%
  mutate(
    ed_primary = ifelse(ed3 == 1, 1, 0),
    ed_secondary = ifelse(ed3 == 2, 1, 0),
    ed_tertiary = ifelse(ed3 == 3, 1, 0),
    missing = case_when(is.na(skill_reading) | is.na(NFE12) | is.na(FORBORN) | is.na(children) | is.na(FORLANG) | is.na(ed3) | is.na(ed_currently) ~ 1,
                        .default = NA)
  ) %>%
  select(-ed3) %>%
  select(iso3c, skill_reading, NFE12, FORBORN, children, FORLANG, ed_primary, ed_secondary,
ed_tertiary, missing, aweight)

# drop countries that do not have all variables
valid_countries <- table_1 %>%
  group_by(iso3c) %>%
  summarise(across(everything(), ~ sum(!is.na(.)))) %>%
  filter(across(-iso3c, ~ . > 0)) %>%
  pull(iso3c)
```

```

table_1 <- table_1 %>%
  filter(iso3c %in% valid_countries)

table_lout <- table_1 %>%
  select(-iso3c) %>%
  summarise(across(
    everything(),
    list(
      mean = ~ mean(., na.rm = TRUE),
      wmean = ~ weighted.mean(., aweight, na.rm = TRUE),
      sd = ~ sd(., na.rm = TRUE),
      min = ~ min(., na.rm = TRUE),
      max = ~ max(., na.rm = TRUE),
      n = ~ sum(!is.na(.))
    )
  ),
  .names = "{.col}__{.fn}"
) %>%
pivot_longer(
  everything(),
  names_to = c("variable", ".value"),
  names_sep = "__"
) %>%
mutate(variable = case_when(
  variable == "skill_reading" ~ "Literacy Skills",
  variable == "NFE12" ~ "ALE Participation",
  variable == "FORBORN" ~ "Foreign-Born",
  variable == "children" ~ "Has Children",
  variable == "FORLANG" ~ "Non-Native Speaker",
  variable == "ed_primary" ~ "Primary Education",
  variable == "ed_secondary" ~ "Secondary Education",
  variable == "ed_tertiary" ~ "Tertiary Education",
  variable == "missing" ~ "Listwise Missing Cases",
  TRUE ~ variable
))

# Save as HTML
table_lout %>%
  select(variable, mean, wmean, sd, min, max, n) %>%
  kable("html", digits = 2, caption = "Descriptive Statistics with Weighted Means") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  save_kable(here("Results", "table_S1.html"))

table_lout %>% kable("html", digits = 2, caption = "Descriptive Statistics") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
write.csv(
  table_lout,
  file = here("Results", "TS1f.csv"),
  row.names = FALSE
)

```

Descriptive Statistics

variable	mean	wmean	sd	min	max	n
Literacy Skills	256.83	260.91	53.74	23.83	420.67	67669
ALE Participation	0.42	0.47	0.49	0.00	1.00	66807
Foreign-Born	0.13	0.25	0.34	0.00	1.00	66094
Has Children	0.73	0.69	0.45	0.00	1.00	67669
Non-Native Speaker	0.10	0.20	0.31	0.00	1.00	66094
Primary Education	0.12	0.13	0.32	0.00	1.00	66542
Secondary Education	0.41	0.39	0.49	0.00	1.00	66542
Tertiary Education	0.47	0.49	0.50	0.00	1.00	66542
Listwise Missing Cases	1.00	1.00	0.00	1.00	1.00	1864
aweight	22.79	497.71	104.03	1.00	3988.08	67669

5.3 Men

```

table_1a <- df2 %>%
  subset(female == 0 & !is.na(female) & ed_currently != 1) %>%
  mutate(
    ed_primary = ifelse(ed3 == 1, 1, 0),
    ed_secondary = ifelse(ed3 == 2, 1, 0),
    ed_tertiary = ifelse(ed3 == 3, 1, 0),
    missing = case_when(is.na(skill_reading) | is.na(NFE12) | is.na(FORBORN) | is.na(children) | is.na(FORLANG) | is.na(ed3) | is.na(ed_currently) ~ 1,
                        .default = NA)
  ) %>%
  select(-ed3) %>%
  select(iso3c, skill_reading, NFE12, FORBORN, children, FORLANG, ed_primary, ed_secondary,
ed_tertiary, missing, aweight)

# drop countries that do not have all variables
valid_countries <- table_1a %>%
  group_by(iso3c) %>%
  summarise(across(everything(), ~ sum(!is.na(.)))) %>%
  filter(across(-iso3c, ~ . > 0)) %>%
  pull(iso3c)

table_1a <- table_1a %>%
  filter(iso3c %in% valid_countries)

table_louta <- table_1a %>%
  select(-iso3c) %>%

```

```

summarise(across(
  everything(),
  list(
    mean = ~ mean(., na.rm = TRUE),
    wmean = ~ weighted.mean(., aweight, na.rm = TRUE),
    sd = ~ sd(., na.rm = TRUE),
    min = ~ min(., na.rm = TRUE),
    max = ~ max(., na.rm = TRUE),
    n = ~ sum(!is.na(.))
  ),
  .names = "{.col}__{.fn}"
)) %>%
pivot_longer(
  everything(),
  names_to = c("variable", ".value"),
  names_sep = "__"
) %>%
mutate(variable = case_when(
  variable == "skill_reading" ~ "Literacy Skills",
  variable == "NFE12" ~ "ALE Participation",
  variable == "FORBORN" ~ "Foreign-Born",
  variable == "children" ~ "Has Children",
  variable == "FORLANG" ~ "Non-Native Speaker",
  variable == "ed_primary" ~ "Primary Education",
  variable == "ed_secondary" ~ "Secondary Education",
  variable == "ed_tertiary" ~ "Tertiary Education",
  variable == "missing" ~ "Listwise Missing Cases",
  TRUE ~ variable
))

# Save as HTML
table_louta %>%
  select(variable, mean, wmean, sd, min, max, n) %>%
  kable("html", digits = 2, caption = "Descriptive Statistics with Weighted Means") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  save_kable(here("Results", "table_S1.html"))

table_louta %>% kable("html", digits = 2, caption = "Descriptive Statistics") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))

write.csv(
  table_louta,
  file = here("Results", "TS1m.csv"),
  row.names = FALSE
)

```

Descriptive Statistics

variable	mean	wmean	sd	min	max	n
Literacy Skills	255.86	260.87	57.22	16.76	449.50	58763
ALE Participation	0.39	0.46	0.49	0.00	1.00	57862
Foreign-Born	0.13	0.24	0.33	0.00	1.00	57098
Has Children	0.61	0.59	0.49	0.00	1.00	58763
Non-Native Speaker	0.10	0.18	0.31	0.00	1.00	57098
Primary Education	0.14	0.15	0.34	0.00	1.00	57599
Secondary Education	0.47	0.42	0.50	0.00	1.00	57599
Tertiary Education	0.40	0.42	0.49	0.00	1.00	57599
Listwise Missing Cases	1.00	1.00	0.00	1.00	1.00	1927
aweight	26.54	489.01	110.79	1.00	3174.53	58763

5.4 Foreign-born, non-native

```
# Helper function to summarize a group
summarize_group <- function(data) {
  data %>%
    mutate(
      ed_primary = ifelse(ed3 == 1, 1, 0),
      ed_secondary = ifelse(ed3 == 2, 1, 0),
      ed_tertiary = ifelse(ed3 == 3, 1, 0)
    ) %>%
    summarise(across(
      c(skill_reading, NFE12, FORBORN, children, FORLANG,
        ed_primary, ed_secondary, ed_tertiary, ed_currently),
      list(
        mean = ~ mean(.x, na.rm = TRUE),
        wmean = ~ weighted.mean(.x, aweight, na.rm = TRUE)
      ),
      .names = "{.col}__{.fn}"
    )) %>%
    pivot_longer(
      everything(),
      names_to = c("variable", ".value"),
      names_sep = "__"
    )
}

# Summarize females
female_sum <- df2 %>%
  filter(female == 1, FORLANG == 1, FORBORN == 1) %>%
```

```

    summarize_group()

# Summarize males
male_sum <- df2 %>%
  filter(female == 0, FORLANG == 1, FORBORN == 1) %>%
  summarize_group()

# Combine both into one table
table_summary <- female_sum %>%
  rename(female_mean = mean, female_wmean = wmean) %>%
  full_join(
    male_sum %>%
      rename(male_mean = mean, male_wmean = wmean),
    by = "variable"
  ) %>%
  mutate(variable = case_when(
    variable == "skill_reading" ~ "Literacy Skills",
    variable == "NFE12" ~ "ALE Participation",
    variable == "FORBORN" ~ "Foreign-Born",
    variable == "children" ~ "Has Children",
    variable == "FORLANG" ~ "Non-Native Speaker",
    variable == "ed_currently" ~ "Currently in Education",
    variable == "ed_primary" ~ "Primary Education",
    variable == "ed_secondary" ~ "Secondary Education",
    variable == "ed_tertiary" ~ "Tertiary Education",
    TRUE ~ variable
  ))

# Create formatted table
table_output <- table_summary %>%
  kable("html", digits = 2, caption = "Means and Weighted Means by Gender") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))

# Save to HTML file
save_kable(table_output, file = here::here("Results", "Table_Slag.html"))

# Print table to viewer
table_output

```

Means and Weighted Means by Gender

variable	female_mean	female_wmean	male_mean	male_wmean
Literacy Skills	236.16	247.32	234.02	249.38
ALE Participation	0.42	0.52	0.42	0.53
Foreign-Born	1.00	1.00	1.00	1.00
Has Children	0.68	0.66	0.59	0.57
Non-Native Speaker	1.00	1.00	1.00	1.00
Primary Education	0.18	0.12	0.22	0.15
Secondary Education	0.32	0.29	0.33	0.26
Tertiary Education	0.50	0.59	0.45	0.59
Currently in Education	0.13	0.18	0.13	0.18

5.5 Table S6. Omitted Intercategories

```
summary_table <- df2 %>%
  mutate(
    g1 = ifelse(FORBORN == 0 & FORLANG == 1, 1, 0),
    g1_ale = ifelse(g1 == 1 & NFE12 == 1, 1, 0),

    g2 = ifelse(FORBORN == 1 & FORLANG == 0, 1, 0),
    g2_ale = ifelse(g2 == 1 & NFE12 == 1, 1, 0)
  ) %>%
  group_by(iso3c) %>%

  summarise(
    # Native-born, non-native speaker
    pct_g1 = weighted.mean(g1, SPFWT0, na.rm = TRUE) * 100,
    pct_g1_ale = weighted.mean(g1_ale, SPFWT0, na.rm = TRUE) * 100,
    ratio_g1 = pct_g1 / pct_g1_ale,

    # Foreign-born, native speaker
    pct_g2 = weighted.mean(g2, SPFWT0, na.rm = TRUE) * 100,
    pct_g2_ale = weighted.mean(g2_ale, SPFWT0, na.rm = TRUE) * 100,
    ratio_g2 = pct_g2 / pct_g2_ale,

    .groups = "drop"
  ) %>%
  # Country population
  left_join(
    wb_data("SP.POP.TOTL", start_date = 2023, end_date = 2023) %>%
      select(iso3c, pop_2023 = SP.POP.TOTL),
    by = "iso3c"
  ) %>%
```

```

select(iso3c, pop_2023, pct_g1, pct_g1_ale, ratio_g1, pct_g2, pct_g2_ale, ratio_g2
)

summary_table <- summary_table %>%
  mutate(across(c(starts_with("pct_"), starts_with("ratio_")), ~ round(.x, 2)))

kable(
  summary_table,
  col.names = c("ISO3", "Pop 2023",
    "Nat-born/Non-nat (%)", "ALE Nat-born/Non-nat (%)", "Ratio",
    "FB/Nat-speaker (%)", "ALE FB/Nat-speaker (%)", "Ratio")
)

write.csv(
  summary_table,
  file = here("Results", "TS2.csv"),
  row.names = FALSE
)

```

ISO3	Pop 2023	Nat-born/Non-nat (%)	ALE Nat-born/Non-nat (%)	Ratio	FB/Nat-speaker (%)	ALE FB/Nat-speaker (%)	Ratio
AUT	9131761	2.54	0.85	3.01	5.17	2.14	2.42
BEL	11779946	2.82	1.02	2.75	3.84	1.53	2.51
CAN	40083484	4.26	1.93	2.21	11.33	6.40	1.77
CHE	8888822	4.83	1.96	2.46	15.36	7.19	2.14
CHL	19658835	0.22	0.07	3.15	8.10	2.59	3.13
CZE	10864042	0.36	0.13	2.72	1.31	0.52	2.51
DEU	83287273	2.32	0.80	2.90	3.08	1.08	2.84
ESP	48352528	3.17	1.23	2.57	13.61	5.41	2.52
EST	1370286	1.69	0.84	2.02	8.63	3.76	2.30
FIN	5583911	0.98	0.61	1.60	1.58	0.69	2.29
FRA	68372286	1.50	0.51	2.96	5.99	1.78	3.36
GBR	68492000	1.76	0.83	2.13	7.56	4.22	1.79
HRV	3859686	0.68	0.05	13.53	6.91	1.49	4.63
HUN	9592186	0.64	0.13	5.12	3.67	1.25	2.94
IRL	5311538	0.55	0.17	3.14	14.55	8.20	1.78
ISR	9849000	4.42	1.52	2.91	3.61	1.10	3.29
ITA	58984216	2.42	0.80	3.01	3.47	0.73	4.75
JPN	124516650	0.01	0.00	Inf	0.42	0.18	2.30
KOR	51712619	0.13	0.01	12.44	1.41	0.10	13.58
LTU	2871585	4.73	1.32	3.57	1.71	0.61	2.80
LVA	1883710	6.31	2.92	2.16	6.86	2.77	2.47
NOR	5519601	0.51	0.32	1.58	0.39	0.34	1.15
POL	36687353	0.33	0.10	3.12	0.19	0.00	Inf
PRT	10578174	0.69	0.26	2.63	19.65	6.32	3.11
SGP	5917648	36.57	14.90	2.45	7.18	3.92	1.83
SVK	5426740	2.57	0.69	3.71	0.75	0.20	3.72
SWE	10536632	1.26	0.68	1.87	1.69	0.89	1.89
USA	336806231	4.56	2.28	2.00	5.46	3.37	1.62

6 Fig 1. Intersectionality of Literacy

6.1 Define all groups

6.1.1 All countries

```
# helper defined in IRT_helper_groups.R
calc_groups_pv(
  df1_rep = df1_rep,
  df2_rep = df2_rep,
  pvlit    = pvlit,
  pvnum    = NA,
  group_vars = c("ALL", "male", "female", "male_not", "male_fb", "female_not", "female_fb",
    "male_not_natlang", "male_not_notnatlang", "male_fb_natlang", "male_fb_notnatlang", "female_not_natlang",
    "female_not_notnatlang", "female_fb_natlang", "female_fb_notnatlang", "male_not_natlang_child",
    "male_not_notnatlang_child", "male_fb_natlang_child", "male_fb_notnatlang_child", "male_not_natlang_nochild",
    "male_not_notnatlang_nochild", "male_fb_natlang_nochild", "male_fb_notnatlang_nochild", "female_not_natlang_child",
    "female_not_notnatlang_child", "female_fb_natlang_child", "female_fb_notnatlang_child", "female_not_natlang_nochild",
    "female_not_notnatlang_nochild", "female_fb_natlang_nochild", "female_fb_notnatlang_nochild")
)

# combine all mean and se objects into a df
source(here("Sub", "combine_g_objects.R"))

# combine all pct of groups with PIAAC level at 1 or below
source(here("Sub", "combine_g_objects_pct.R"))

# remove raw objects
rm(list = ls(pattern = "^g[0-9]+_(mean|se)_(lit|num)_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_(mean|se)_(lit|num)_diff$" ))
rm(list = ls(pattern = "^g[0-9]+_(pct)_lit01_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_ale_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_sec_df[12]$" ))

# pct is literacy-only so far (num can be calculated the same way)

# give unique naming (note that ALE is already included here so we don't need a new df for it)
df_lit_all <- df_lit
df_num_all <- df_num
df_pct_level_1_or_less_all <- df_pct_level_1_or_less
```

6.1.2 Rich countries only

These are destination countries

```
calc_groups_pv(
  df1_rep = df1_rep_rich,
  df2_rep = df2_rep_rich,
  pvlit    = pvlit,
  pvnum    = NA,
  group_vars = c("ALL", "male", "female", "male_not", "male_fb", "female_not", "female_fb",
    "male_not_natlang", "male_not_notnatlang", "male_fb_natlang", "male_fb_notnatlang", "female_not_natlang",
    "female_not_notnatlang", "female_fb_natlang", "female_fb_notnatlang", "male_not_natlang_child",
    "male_not_notnatlang_child", "male_fb_natlang_child", "male_fb_notnatlang_child", "male_not_natlang_nochild",
    "male_not_notnatlang_nochild", "male_fb_natlang_nochild", "male_fb_notnatlang_nochild")
)
```

```

_nochild", "male_fb_notnatlang_nochild", "female_not_natlang_child", "female_not_notnatlang_child", "female_fb_natlang_child", "female_fb_notnatlang_child", "female_not_natlang_nochild", "female_not_notnatlang_nochild", "female_fb_natlang_nochild", "female_fb_notnatlang_nochild")
)

source(here("Sub", "combine_g_objects.R"))

# PIAAC level at 1 or below
source(here("Sub", "combine_g_objects_pct.R"))

# remove raw objects
rm(list = ls(pattern = "^g[0-9]+_(mean|se)_lit_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_(pct|se)_lit01_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_ale_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_sec_df[12]$" ))

# unique names
df_lit_rich <- df_lit
df_pct_level_1_or_less_rich <- df_pct_level_1_or_less

```

6.1.3 Europe only

```

calc_groups_pv(
  df1_rep = df1_rep_eur,
  df2_rep = df2_rep_eur,
  pvlit   = pvlit,
  pvnum   = NA,
  group_vars = c("ALL", "male", "female", "male_not", "male_fb", "female_not", "female_fb", "male_not_natlang", "male_not_notnatlang", "male_fb_natlang", "male_fb_notnatlang", "female_not_natlang", "female_not_notnatlang", "female_fb_natlang", "female_fb_notnatlang", "male_not_natlang_child", "male_not_notnatlang_child", "male_fb_natlang_child", "male_fb_notnatlang_child", "male_not_natlang_nochild", "male_not_notnatlang_nochild", "male_fb_natlang_nochild", "male_fb_notnatlang_nochild", "female_not_natlang_child", "female_not_notnatlang_child", "female_fb_natlang_child", "female_fb_notnatlang_child", "female_not_natlang_nochild", "female_not_notnatlang_nochild", "female_fb_natlang_nochild", "female_fb_notnatlang_nochild")
)

source(here("Sub", "combine_g_objects.R"))
source(here("Sub", "combine_g_objects_pct.R"))

rm(list = ls(pattern = "^g[0-9]+_(mean|se)_lit_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_(pct|se)_lit01_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_ale_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_sec_df[12]$" ))

df_lit_eur <- df_lit
df_pct_level_1_or_less_eur <- df_pct_level_1_or_less

```

6.1.4 Germany only

This is for a blog post related to the paper

```

group_vars = c("ALL", "male", "female", "male_not", "male_fb", "female_not", "female_fb", "male_not_natlang", "male_not_notnatlang", "male_fb_natlang", "male_fb_notnatlang", "female_not_natlang", "female_not_notnatlang", "female_fb_natlang", "female_fb_notnatlang", "male_not_natlang_child", "male_not_notnatlang_child", "male_fb_natlang_child", "male_fb_notnatlang_child", "male_not_natlang_nochild", "male_not_notnatlang_nochild", "male_fb_natlang_nochild", "male_fb_notnatlang_nochild")

```

```
ld", "male_fb_notnatlang_nochild", "female_not_natlang_child", "female_not_notnatlang_child",
", "female_fb_natlang_child", "female_fb_notnatlang_child", "female_not_natlang_nochild", "
female_not_notnatlang_nochild", "female_fb_natlang_nochild", "female_fb_notnatlang_nochild"
)

calc_groups_pv(
  df1_rep = df1_rep_de,
  df2_rep = df2_rep_de,
  pvlit    = pvlit,
  pvnum    = pvnum,
  group_vars = group_vars)

source(here("Sub", "combine_g_objects.R"))
source(here("Sub", "combine_g_objects_pct.R"))

rm(list = ls(pattern = "^g[0-9]+_(mean|se)_(lit|num)_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_(mean|se)_(lit|num)_diff$" ))
rm(list = ls(pattern = "^g[0-9]+_(pct|se)_lit01_df[12]$" ))
rm(list = ls(pattern = "^g[0-9]+_pct_ale_df[12]$" ))

df_lit_de <- df_lit
df_num_de <- df_num
df_pct_level_1_or_less_de <- df_pct_level_1_or_less

rm(df_lit, df_num, df_pct_level_1_or_less)
```

6.2 Plot

6.2.1 Prep dfs

```
# function for mapping viridis onto different scalings, e.g., literacy versus percedntages
map_viridis <- function(x) {
  pal <- viridis_pal()(100) # Get 100 viridis colors
  # Cut the numeric vector into 100 bins and map to palette
  idx <- as.numeric(cut(x, breaks = 100))

  res <- rep(NA_character_, length(x))
  res[!is.na(idx)] <- pal[idx[!is.na(idx)]]
  return(res)
}

# All
df_a2 <- df_lit_all %>%
  filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
  select(group_label, mean, pct_ale, pct_sec) %>%
  drop_na(mean)

# Rich
df_a2_rich <- df_lit_rich %>%
  filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
  select(group_label, mean, pct_ale, pct_sec) %>%
  drop_na(mean)

# Europe
df_a2_eur <- df_lit_eur %>%
  filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
  select(group_label, mean, pct_ale, pct_sec) %>%
  drop_na(mean)

# DE
df_a2_de <- df_lit_de %>%
```

```

filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
select(group_label, mean, pct_ale, pct_sec) %>%
drop_na(mean)

df_a2num_de <- df_num_de %>%
  filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
  select(group_label, mean, pct_ale, pct_sec) %>%
  drop_na(mean)

# ChatGPT made this cool hierarchy routine
parent_of <- function(label) {
  if (label == "ALL") return(NA_character_)
  if (!str_detect(label, "_")) return("ALL")
  sub("_[^_]+$", "", label)
}

depth_of <- function(label) {
  if (label == "ALL") return(0L)
  str_count(label, "_") + 1L
}

df_a2 <- df_a2 %>%
  mutate(
    depth = vapply(group_label, depth_of, integer(1)),
    parent = vapply(group_label, parent_of, character(1))
  )

df_a2_rich <- df_a2_rich %>%
  mutate(
    depth = vapply(group_label, depth_of, integer(1)),
    parent = vapply(group_label, parent_of, character(1))
  )

df_a2_eur <- df_a2_eur %>%
  mutate(
    depth = vapply(group_label, depth_of, integer(1)),
    parent = vapply(group_label, parent_of, character(1))
  )

df_a2_de <- df_a2_de %>%
  mutate(
    depth = vapply(group_label, depth_of, integer(1)),
    parent = vapply(group_label, parent_of, character(1))
  )

df_a2num_de <- df_a2num_de %>%
  mutate(
    depth = vapply(group_label, depth_of, integer(1)),
    parent = vapply(group_label, parent_of, character(1))
  )

# make a label map vector instead of case_match each time

plot_label_map <- c(
  "ALL" = "Entire Population",
  "male" = "Male",
  "female" = "Female",
  "male_not" = "Native-born",
  "female_not" = "Native-born",
  "male_fb" = "Foreign-born",
  "female_fb" = "Foreign-born",
  "male_not_natlang" = "Native",
  "male_fb_natlang" = "Native",
  "female_not_natlang" = "Native",

```

```

"female_fb_natlang" = "Native",
"male_not_notnatlang" = "Non-native",
"male_fb_notnatlang" = "Non-native",
"female_not_notnatlang" = "Non-native",
"female_fb_notnatlang" = "Non-native",
"male_not_natlang_child" = "Parent",
"male_fb_natlang_child" = "Parent",
"female_not_natlang_child" = "Parent",
"female_fb_natlang_child" = "Parent",
"male_not_notnatlang_child" = "Parent",
"male_fb_notnatlang_child" = "Parent",
"female_not_notnatlang_child" = "Parent",
"female_fb_notnatlang_child" = "Parent",
"male_not_natlang_nochild" = "Childless",
"male_fb_natlang_nochild" = "Childless",
"female_not_natlang_nochild" = "Childless",
"female_fb_natlang_nochild" = "Childless",
"male_not_notnatlang_nochild" = "Childless",
"male_fb_notnatlang_nochild" = "Childless",
"female_not_notnatlang_nochild" = "Childless",
"female_fb_notnatlang_nochild" = "Childless"
)

# Add group labels for plotting
df_a2 <- df_a2 %>%
  mutate(plot_label = plot_label_map[group_label])

df_a2_rich <- df_a2_rich %>%
  mutate(plot_label = plot_label_map[group_label])

df_a2_eur <- df_a2_eur %>%
  mutate(plot_label = plot_label_map[group_label])

df_a2_de <- df_a2_de %>%
  mutate(plot_label = plot_label_map[group_label])

df_a2num_de <- df_a2num_de %>%
  mutate(plot_label = plot_label_map[group_label])

# organize data for plotting
labels <- df_a2$group_label
#labels <- df_a2_de$group_label

children_map <- df_a2 %>%
  select(group_label, parent) %>%
  filter(!is.na(parent), parent %in% labels) %>%
  arrange(parent, group_label) %>%
  group_by(parent) %>%
  summarise(children = list(group_label), .groups = "drop") %>%
  deframe()

children_map_rich <- df_a2_rich %>%
  select(group_label, parent) %>%
  filter(!is.na(parent), parent %in% labels) %>%
  arrange(parent, group_label) %>%
  group_by(parent) %>%
  summarise(children = list(group_label), .groups = "drop") %>%
  deframe()

children_map_eur <- df_a2_eur %>%
  select(group_label, parent) %>%
  filter(!is.na(parent), parent %in% labels) %>%
  arrange(parent, group_label) %>%

```

```

group_by(parent) %>%
summarise(children = list(group_label), .groups = "drop") %>%
deframe()

children_map_de <- df_a2_de %>%
  select(group_label, parent) %>%
  filter(!is.na(parent), parent %in% labels) %>%
  arrange(parent, group_label) %>%
  group_by(parent) %>%
  summarise(children = list(group_label), .groups = "drop") %>%
  deframe()

children_map_de_num <- df_a2num_de %>%
  select(group_label, parent) %>%
  filter(!is.na(parent), parent %in% labels) %>%
  arrange(parent, group_label) %>%
  group_by(parent) %>%
  summarise(children = list(group_label), .groups = "drop") %>%
  deframe()

# Ensure every label is a key (even if it has no children)
for (lab in labels) if (is.null(children_map[[lab]])) children_map[[lab]] <- character(0)

for (lab in labels) if (is.null(children_map_rich[[lab]])) children_map_rich[[lab]] <- character(0)

for (lab in labels) if (is.null(children_map_eur[[lab]])) children_map_eur[[lab]] <- character(0)

for (lab in labels) if (is.null(children_map_de[[lab]])) children_map_de[[lab]] <- character(0)

for (lab in labels) if (is.null(children_map_de_num[[lab]])) children_map_de_num[[lab]] <- character(0)

max_depth <- max(df_a2$depth, na.rm = TRUE)
max_depth_rich <- max(df_a2_rich$depth, na.rm = TRUE)
max_depth_eur <- max(df_a2_eur$depth, na.rm = TRUE)
max_depth_de <- max(df_a2_de$depth, na.rm = TRUE)
max_depth_de_num <- max(df_a2num_de$depth, na.rm = TRUE)

# Assign x recursively
x_pos <- setNames(rep(NA_real_, length(labels)), labels)

assign_positions <- function(node, depth, x) {
  x_pos[[node]] <- x
  ch <- children_map[[node]]
  if (length(ch) == 2) {
    # symmetric left/right offset that shrinks with depth
    offset <- 2^(max_depth - depth - 1)
    assign_positions(ch[1], depth + 1, x - offset)
    assign_positions(ch[2], depth + 1, x + offset)
  } else if (length(ch) == 1) {
    # single child straight below (although this case doesn't exist in these groupings)
    assign_positions(ch[1], depth + 1, x)
  }
}

assign_positions_rich <- function(node, depth, x) {
  x_pos[[node]] <- x
  ch <- children_map_rich[[node]]
  if (length(ch) == 2) {
    # symmetric left/right offset that shrinks with depth

```

```

    offset <- 2^(max_depth_rich - depth - 1)
    assign_positions_rich(ch[1], depth + 1, x - offset)
    assign_positions_rich(ch[2], depth + 1, x + offset)
  } else if (length(ch) == 1) {
    # single child straight below (although this case doesn't exist in these groupings)
    assign_positions_rich(ch[1], depth + 1, x)
  }
}

assign_positions_eur <- function(node, depth, x) {
  x_pos[[node]] <- x
  ch <- children_map_eur[[node]]
  if (length(ch) == 2) {
    # symmetric left/right offset that shrinks with depth
    offset <- 2^(max_depth_eur - depth - 1)
    assign_positions_eur(ch[1], depth + 1, x - offset)
    assign_positions_eur(ch[2], depth + 1, x + offset)
  } else if (length(ch) == 1) {
    # single child straight below (although this case doesn't exist in these groupings)
    assign_positions_eur(ch[1], depth + 1, x)
  }
}

# DE
assign_positions_de <- function(node, depth, x) {
  x_pos[[node]] <- x
  ch <- children_map_de[[node]]
  if (length(ch) == 2) {
    offset <- 2^(max_depth_de - depth - 1)
    assign_positions_de(ch[1], depth + 1, x - offset)
    assign_positions_de(ch[2], depth + 1, x + offset)
  } else if (length(ch) == 1) {
    assign_positions_de(ch[1], depth + 1, x)
  }
}

assign_positions_de_num <- function(node, depth, x) {
  x_pos[[node]] <- x
  ch <- children_map_de_num[[node]]
  if (length(ch) == 2) {
    offset <- 2^(max_depth_de_num - depth - 1)
    assign_positions_de_num(ch[1], depth + 1, x - offset)
    assign_positions_de_num(ch[2], depth + 1, x + offset)
  } else if (length(ch) == 1) {
    assign_positions_de_num(ch[1], depth + 1, x)
  }
}

root <- if ("ALL" %in% labels) "ALL" else df_a2 %>% arrange(depth) %>% slice(1) %>% pull(group_label)
assign_positions(root, 0L, 0)

root_rich <- if ("ALL" %in% labels) "ALL" else df_a2_rich %>% arrange(depth) %>% slice(1) %>% pull(group_label)
assign_positions_rich(root_rich, 0L, 0)

root_eur <- if ("ALL" %in% labels) "ALL" else df_a2_eur %>% arrange(depth) %>% slice(1) %>% pull(group_label)
assign_positions_eur(root_eur, 0L, 0)

root_de <- if ("ALL" %in% labels) "ALL" else df_a2_de %>% arrange(depth) %>% slice(1) %>% pull(group_label)
assign_positions_de(root_de, 0L, 0)

```

```

root_de_num <- if ("ALL" %in% labels) "ALL" else df_a2num_de %>% arrange(depth) %>% slice(1)
) %>% pull(group_label)
assign_positions_de_num(root_de_num, 0L, 0)

# Top is y=0, then working down (by going up in count)
df_a2 <- df_a2 %>%
  mutate(
    x = unname(x_pos[group_label]),
    y = depth
  )

df_a2_rich <- df_a2_rich %>%
  mutate(
    x = unname(x_pos[group_label]),
    y = depth
  )

df_a2_eur <- df_a2_eur %>%
  mutate(
    x = unname(x_pos[group_label]),
    y = depth
  )

df_a2_de <- df_a2_de %>%
  mutate(
    x = unname(x_pos[group_label]),
    y = depth
  )

df_a2num_de <- df_a2num_de %>%
  mutate(
    x = unname(x_pos[group_label]),
    y = depth
  )

# map colors using new viridis scale function

df_a2 <- df_a2 %>%
  mutate(color_lit = map_viridis(mean), color_ale = map_viridis(pct_ale), color_sec = map_viridis(pct_sec))

df_a2_rich <- df_a2_rich %>%
  mutate(color_lit = map_viridis(mean), color_ale = map_viridis(pct_ale), color_sec = map_viridis(pct_sec))

df_a2_eur <- df_a2_eur %>%
  mutate(color_lit = map_viridis(mean), color_ale = map_viridis(pct_ale), color_sec = map_viridis(pct_sec))

df_a2_de <- df_a2_de %>%
  mutate(color_lit = map_viridis(mean), color_ale = map_viridis(pct_ale), color_sec = map_viridis(pct_sec))

df_a2num_de <- df_a2num_de %>%
  mutate(color_lit = map_viridis(mean), color_ale = map_viridis(pct_ale), color_sec = map_viridis(pct_sec))

#connecting lines
edges <- df_a2 %>%
  #mutate(y = y + 1) %>%
  select(child = group_label, xend = x, yend = y, parent) %>%
  left_join(df_a2 %>% select(parent = group_label, x, y),
    by = "parent", relationship = "many-to-one") %>%
  filter(!is.na(parent))

```



```

edges_rich <- df_a2_rich %>%
  #mutate(y = y + 1) %>%
  select(child = group_label, xend = x, yend = y, parent) %>%
  left_join(df_a2_rich %>% select(parent = group_label, x, y),
            by = "parent", relationship = "many-to-one") %>%
  filter(!is.na(parent))

edges_eur <- df_a2_eur %>%
  #mutate(y = y + 1) %>%
  select(child = group_label, xend = x, yend = y, parent) %>%
  left_join(df_a2_eur %>% select(parent = group_label, x, y),
            by = "parent", relationship = "many-to-one") %>%
  filter(!is.na(parent))

# DE
edges_de <- df_a2_de %>%
  #mutate(y = y + 1) %>%
  select(child = group_label, xend = x, yend = y, parent) %>%
  left_join(df_a2_de %>% select(parent = group_label, x, y),
            by = "parent", relationship = "many-to-one") %>%
  filter(!is.na(parent))

edges_de_num <- df_a2num_de %>%
  #mutate(y = y + 1) %>%
  select(child = group_label, xend = x, yend = y, parent) %>%
  left_join(df_a2num_de %>% select(parent = group_label, x, y),
            by = "parent", relationship = "many-to-one") %>%
  filter(!is.na(parent))

```

6.2.2 Scale adjustment

```

x_sclr <- 3

df_a2 <- df_a2 %>%
  mutate(x = x * x_sclr)

edges <- edges %>%
  mutate(xend = xend * x_sclr,
         x = x * x_sclr)

df_a2_rich <- df_a2_rich %>%
  mutate(x = x * x_sclr)

edges_rich <- edges_rich %>%
  mutate(xend = xend * x_sclr,
         x = x * x_sclr)

df_a2_eur <- df_a2_eur %>%
  mutate(x = x * x_sclr)

edges_eur <- edges_eur %>%
  mutate(xend = xend * x_sclr,
         x = x * x_sclr)

df_a2_de <- df_a2_de %>%
  mutate(x = x * x_sclr)

edges_de <- edges_de %>%
  mutate(xend = xend * x_sclr,
         x = x * x_sclr)

```

6.2.3 A. All

```
agg_png(here("Results", "Fig2A_A.png"), res = 200, width = 2000, height = 1000)

# Petal size parameters
rx <- 0.8 # Horizontal radius
ry <- 0.15 # Vertical radius

# Dynamic Legend Coordinates
# Find far-left and far-right
min_x <- min(df_a2$x, na.rm = TRUE) + 8
max_x <- max(df_a2$x, na.rm = TRUE) - 8

leg_x <- min_x + 1 # Top-left anchor for the flower legend
leg_y <- 0 # Aligned vertically with the root "ALL" node

bar_w <- (max_x - min_x) * 0.15 # Width of the color bar (15% of total plot width)
bar_x <- max_x - bar_w # Top-right anchor for the color bar
bar_y <- 0

p <- ggplot(df_a2, aes(x = x, y = y)) +
  # Connecting Lines
  geom_segment(
    data = edges,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +

  # FLOWER PETALS (DATA)
  # 4 o'clock Petal (ALE)
  geom_segment(
    aes(xend = x + rx * 0.866, yend = y + ry * 0.7, color = color_ale),
    linewidth = 3.5, lineend = "round"
  ) +
  # 8 o'clock Petal (Secondary Ed)
  geom_segment(
    aes(xend = x - rx * 0.866, yend = y + ry * 0.7, color = color_sec),
    linewidth = 3.5, lineend = "round"
  ) +
  # 12 o'clock Petal (Literacy)
  geom_segment(
    aes(xend = x, yend = y - ry, color = color_lit),
    linewidth = 3.5, lineend = "round"
  ) +

  # PETAL LABELS
  geom_text(aes(x = x, y = y - ry - 0.1, label = round(mean, 0)), vjust = 0, hjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x + rx * 0.866 + 0.3, y = y + ry * 0.5 + 0.05, label = paste0(" ", round(pct_ale, 0), c("% ", rep("", 30)))), hjust = 0, vjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x - rx * 0.866 - 0.3, y = y + ry * 0.5 + 0.05, label = paste0(round(pct_sec, 0), c("% ", rep("", 30)))), hjust = 1, vjust = 0.5, size = 2.2, color = "black") +

  # GROUP SPLIT LABELS
  geom_text(aes(label = plot_label), vjust = 5.0, size = 1.9, color = "black") +

  # Top-Left FLOWER LEGEND
  # Grey Petals for the Key
  annotate("segment", x = leg_x, xend = leg_x, y = leg_y, yend = leg_y - ry, linewidth = 3.5, color = "grey60", lineend = "round") +
```

```

    annotate("segment", x = leg_x, xend = leg_x + rx * 0.866, y = leg_y, yend = leg_y + ry *
0.7, linewidth = 3.5, color = "grey60", lineend = "round") +
    annotate("segment", x = leg_x, xend = leg_x - rx * 0.866, y = leg_y, yend = leg_y + ry *
0.7, linewidth = 3.5, color = "grey60", lineend = "round") +
    # Line-break Text Labels
    annotate("text", x = leg_x, y = leg_y - ry - 0.15, label = "Literacy\n(Avg.)", vjust = 0,
hjust = 0.5, size = 2.2, color = "black") +
    annotate("text", x = leg_x + rx * 0.866 + 4.5, y = leg_y + ry * 0.5 + 0.05, label = "ALE
participati\non (%)", hjust = 0.5, vjust = 0.5, size = 2.2, color = "black") +
    annotate("text", x = leg_x - rx * 0.866 - 4.5, y = leg_y + ry * 0.5 + 0.05, label = "Tert
iary\ncompletion (%)", hjust = 0.5, vjust = 0.5, size = 2.2, color = "black") +

    # Top right legend
    # connected rectangles gradient bar
    annotate("rect",
            xmin = bar_x + (0:8) * (bar_w/9),
            xmax = bar_x + (1:9) * (bar_w/9),
            ymin = bar_y + 0.09,
            ymax = bar_y + 0.15,
            fill = viridis_pal()(9)) +
    annotate("text", x = bar_x - 0.3, y = bar_y + 0.12, label = "Lowest values", hjust = 1, v
just = 0.5, size = 2.2, color = "black") +
    annotate("text", x = bar_x + bar_w + 0.3, y = bar_y + 0.12, label = "Highest values", hju
st = 0, vjust = 0.5, size = 2.2, color = "black") +

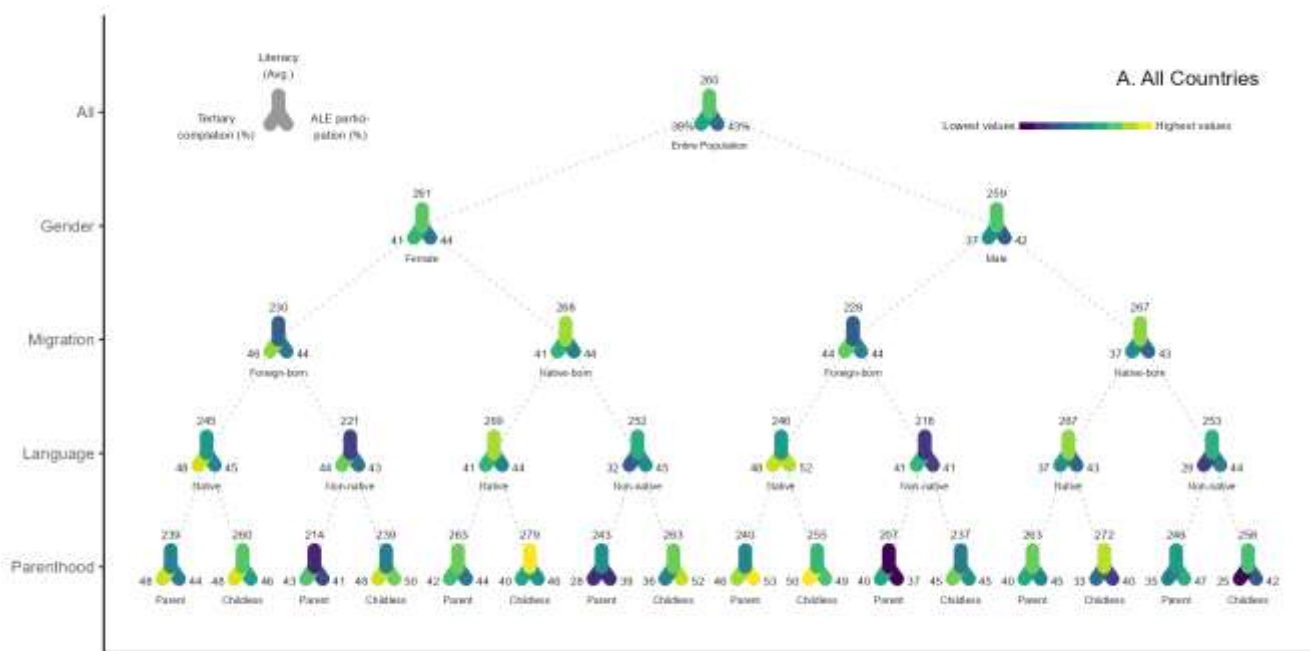
    # Interpret pre-calculated hex literal
    scale_color_identity() +

    # Expand ylimits
    scale_y_reverse(
        breaks = 0:4,
        labels = c("All", "Gender", "Migration", "Language", "Parenthood"),
        limits = c(4.5, -0.6) # Slightly expanded to ensure the top legend labels don't get cut
off
    ) +
    labs(
        x = NULL, y = NULL,
        title = NULL
    ) +
    annotate("text", x = 40, y = -0.3,
            label = "A. All Countries", size = 4) +
    theme_classic() +
    theme(
        legend.position = "none",
        panel.grid.minor = element_blank(),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank()
    )

p

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_A.png"))

```



6.2.4 B. Rich

```
agg_png(here("Results", "Fig2A_B.png"), res = 200, width = 1850, height = 1000)

rx <- 0.8
ry <- 0.15
bar_w <- (max_x - min_x) * 0.15
bar_x <- max_x - bar_w
bar_y <- 0

p <- ggplot(df_a2_rich, aes(x = x, y = y)) +
  geom_segment(
    data = edges_rich,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +
  geom_segment(
    aes(xend = x + rx * 0.866, yend = y + ry * 0.7, color = color_ale),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x - rx * 0.866, yend = y + ry * 0.7, color = color_sec),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x, yend = y - ry, color = color_lit),
    linewidth = 3.5, lineend = "round"
  ) +

  geom_text(aes(x = x, y = y - ry - 0.1, label = round(mean, 0)), vjust = 0, hjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x + rx * 0.866 + 0.3, y = y + ry * 0.5 + 0.05, label = paste0(" ", round(pct_ale, 0), c("% ", rep("", 30)))), hjust = 0, vjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x - rx * 0.866 - 0.3, y = y + ry * 0.5 + 0.05, label = paste0(round(pct_sec, 0), c("% ", rep("", 30)))), hjust = 1, vjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(label = plot_label), vjust = 5.0, size = 1.9, color = "black") +
```

```

scale_color_identity() +
scale_y_reverse(
  breaks = 0:4,
  labels = c("", "", "", "", ""),
  limits = c(4.5, -0.6)
) +
labs(
  x = NULL, y = NULL,
  title = NULL
) +
annotate("text", x = 40, y = -0.3,
  label = "B. Rich", size = 4) +
theme_classic() +
theme(
  legend.position = "none",
  panel.grid.minor = element_blank(),
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank()
)

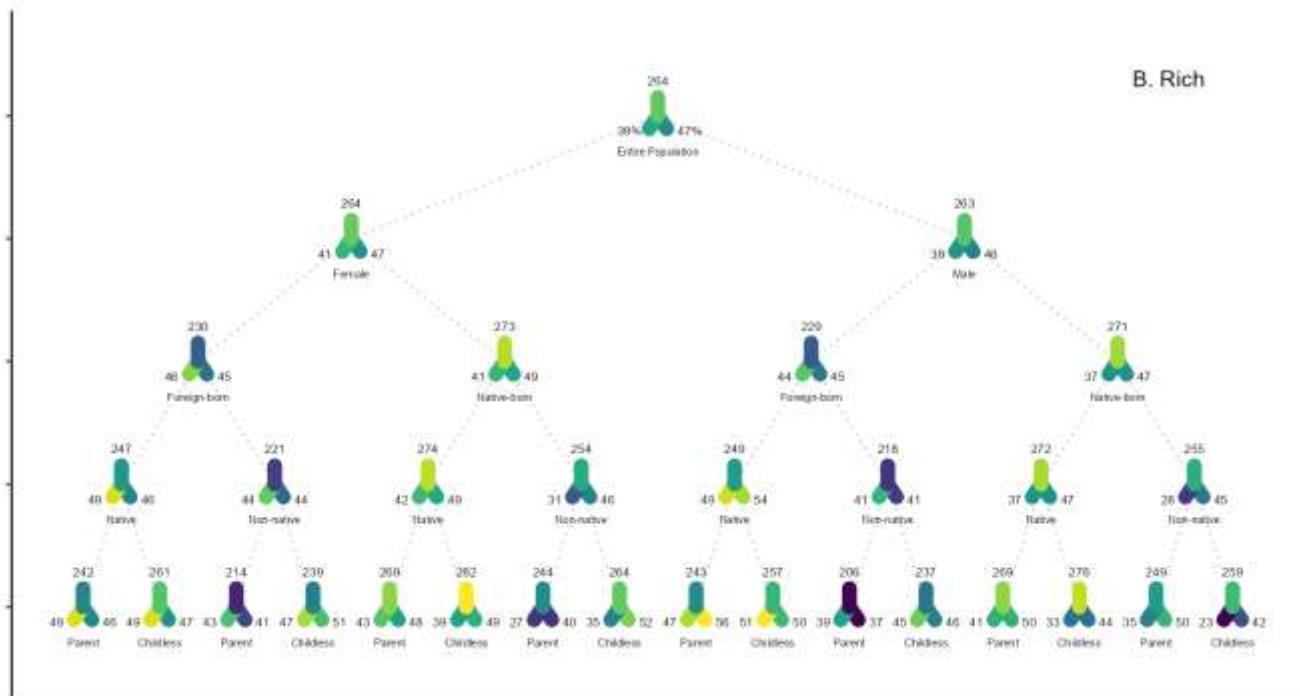
```

p

```

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_B.png"))

```



6.2.5 C. Europe

```

agg_png(here("Results", "Fig2A_C.png"), res = 200, width = 2000, height = 1000)

rx <- 0.8
ry <- 0.15
bar_w <- (max_x - min_x) * 0.15
bar_x <- max_x - bar_w

```

```

bar_y <- 0

p <- ggplot(df_a2_eur, aes(x = x, y = y)) +
  geom_segment(
    data = edges_eur,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +
  geom_segment(
    aes(xend = x + rx * 0.866, yend = y + ry * 0.7, color = color_ale),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x - rx * 0.866, yend = y + ry * 0.7, color = color_sec),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x, yend = y - ry, color = color_lit),
    linewidth = 3.5, lineend = "round"
  ) +

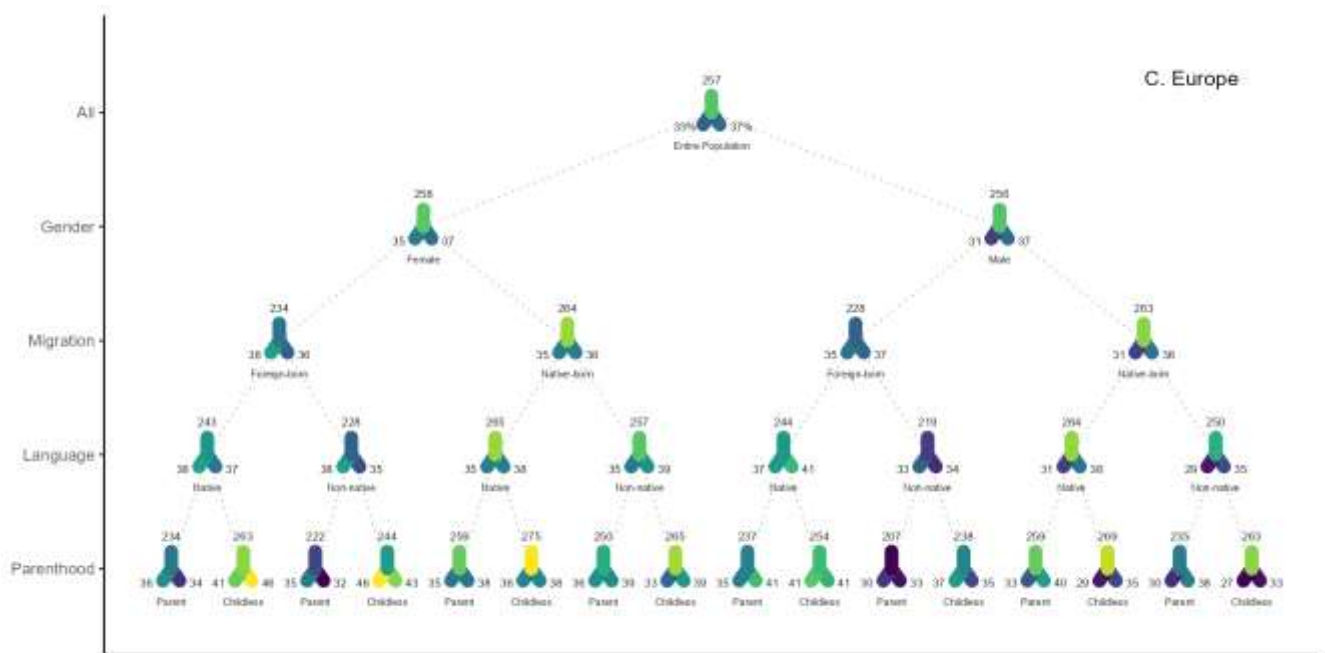
  geom_text(aes(x = x, y = y - ry - 0.1, label = round(mean, 0)), vjust = 0, hjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x + rx * 0.866 + 0.3, y = y + ry * 0.5 + 0.05, label = paste0(" ", round(pct_ale, 0), c("%", rep("", 30)))), hjust = 0, vjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x - rx * 0.866 - 0.3, y = y + ry * 0.5 + 0.05, label = paste0(round(pct_sec, 0), c("% ", rep("", 30)))), hjust = 1, vjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(label = plot_label), vjust = 5.0, size = 1.9, color = "black") +

  scale_color_identity() +
  scale_y_reverse(
    breaks = 0:4,
    labels = c("All", "Gender", "Migration", "Language", "Parenthood"),
    limits = c(4.5, -0.6)
  ) +
  labs(
    x = NULL, y = NULL,
    title = NULL
  ) +
  annotate("text", x = 40, y = -0.3,
    label = "C. Europe", size = 4) +
  theme_classic() +
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )

p

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_C.png"))

```



6.2.6 D. DEU

6.2.6.1 Lit English Labels

```
agg_png(here("Results", "Fig2A_D.png"), res = 200, width = 1850, height = 1000)

rx <- 0.8
ry <- 0.15
bar_w <- (max_x - min_x) * 0.15
bar_x <- max_x - bar_w
bar_y <- 0

p <- ggplot(df_a2_de, aes(x = x, y = y)) +
  geom_segment(
    data = edges_de,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +
  geom_segment(
    aes(xend = x + rx * 0.866, yend = y + ry * 0.7, color = color_ale),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x - rx * 0.866, yend = y + ry * 0.7, color = color_sec),
    linewidth = 3.5, lineend = "round"
  ) +
  geom_segment(
    aes(xend = x, yend = y - ry, color = color_lit),
    linewidth = 3.5, lineend = "round"
  ) +

  geom_text(aes(x = x, y = y - ry - 0.1, label = round(mean, 0)), vjust = 0, hjust = 0.5, size = 2.2, color = "black") +
  geom_text(aes(x = x + rx * 0.866 + 0.3, y = y + ry * 0.5 + 0.05, label = paste0(" ", round(pct_ale, 0), c("%", rep("", 30)))), hjust = 0, vjust = 0.5, size = 2.2, color = "black") +
  +
```

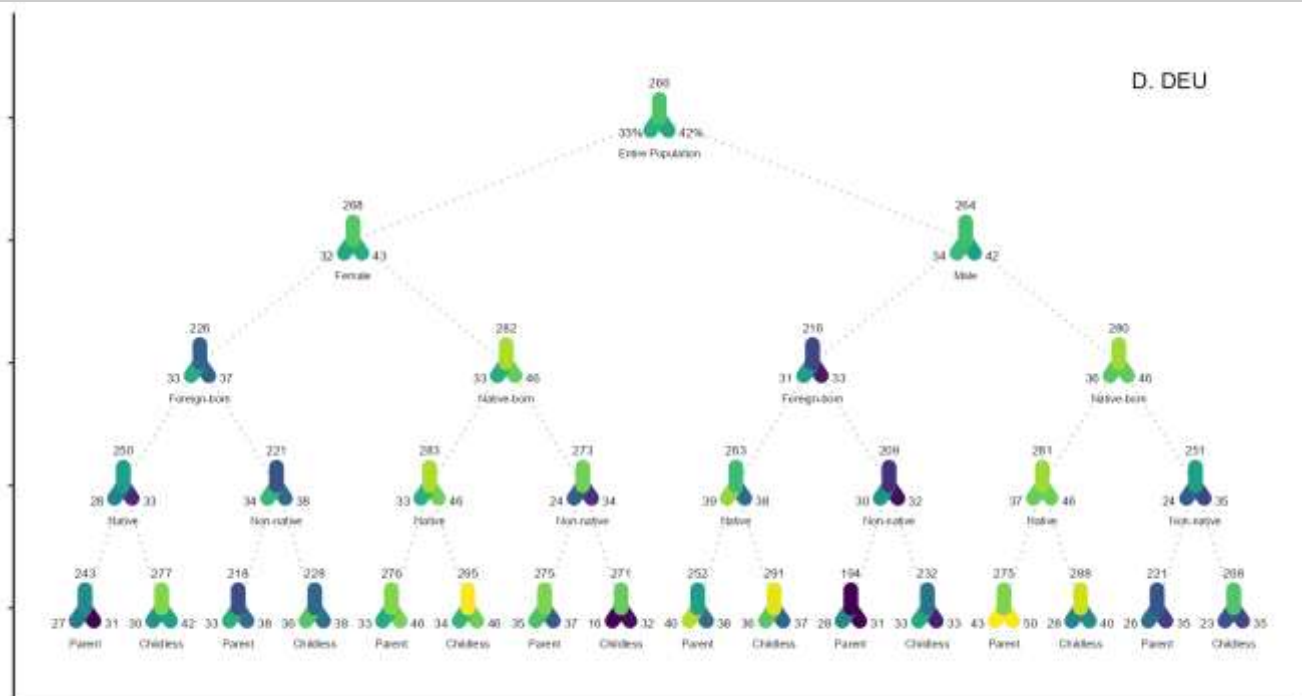
```

    geom_text(aes(x = x - rx * 0.866 - 0.3, y = y + ry * 0.5 + 0.05, label = paste0(round(pct
_sec, 0), c("% ", rep(" ", 30)))), hjust = 1, vjust = 0.5, size = 2.2, color = "black") +
    geom_text(aes(label = plot_label), vjust = 5.0, size = 1.9, color = "black") +

    scale_color_identity() +
    scale_y_reverse(
      breaks = 0:4,
      labels = c("", "", "", "", ""),
      limits = c(4.5, -0.6)
    ) +
    labs(
      x = NULL, y = NULL,
      title = NULL
    ) +
    annotate("text", x = 40, y = -0.3,
      label = "D. DEU", size = 4) +
    theme_classic() +
    theme(
      legend.position = "none",
      panel.grid.minor = element_blank(),
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  p

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_D.png"))

```



6.2.6.2 Lit DE Labels

```

agg_png(here("Results", "Fig2A_D_Deutsch.png"), res = 576, width = 3200, height = 2250)

df_a2_de_X <- df_a2_de %>%

```



```

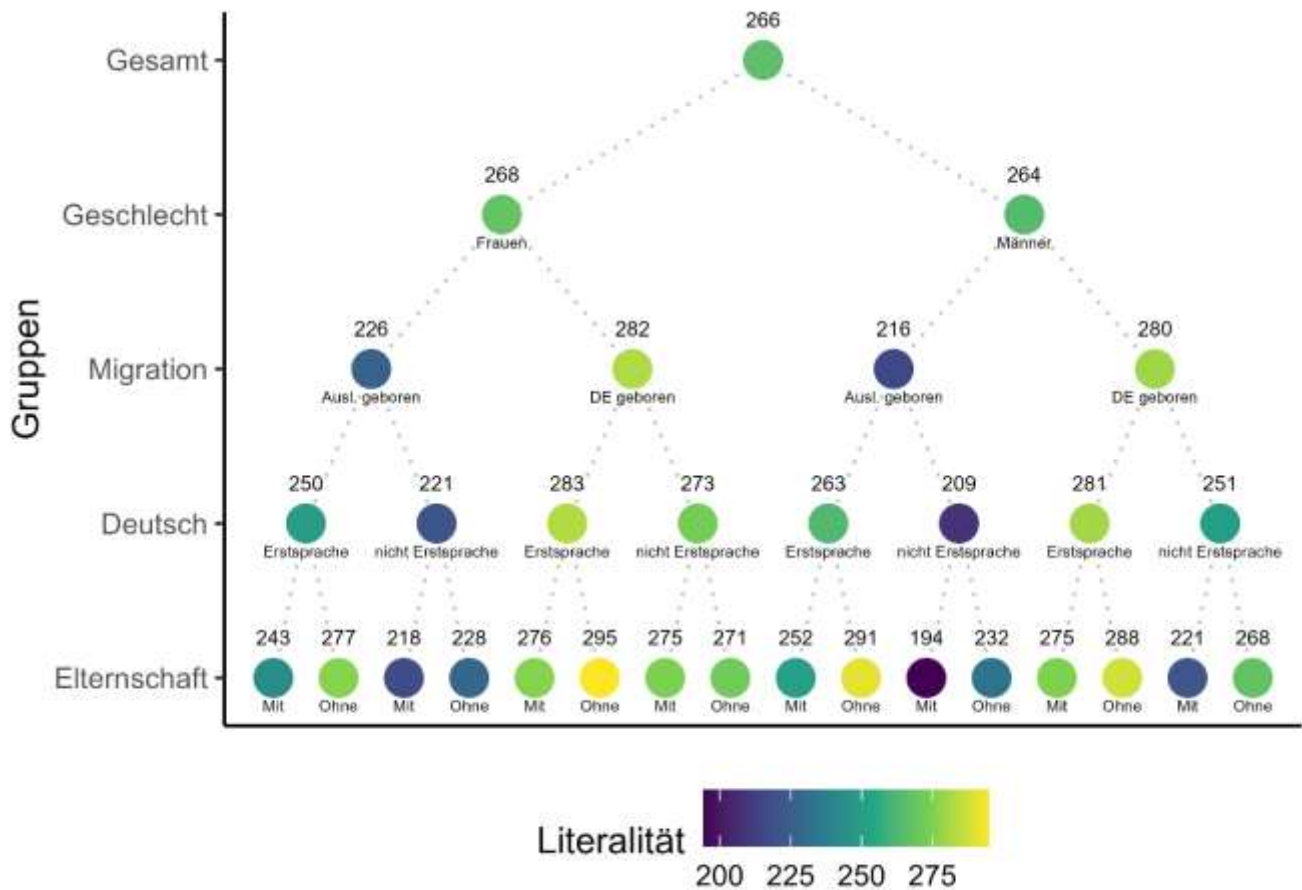
mutate(plot_label = case_match(plot_label,
                                "Male" ~ "Männer",
                                "Female" ~ "Frauen",
                                "Native-born" ~ "DE geboren",
                                "Foreign-born" ~ "Ausl. geboren",
                                "Native" ~ "Erstsprache",
                                "Non-native" ~ "nicht Erstsprache",
                                "Parent" ~ "Mit",
                                "Childless" ~ "Ohne")
)

pdex <- ggplot(df_a2_de_X, aes(x = x, y = y, color = mean)) +
  geom_segment(
    data = edges_de,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +
  geom_point(size = 5) +
  geom_text(
    aes(label = round(mean, 0)),
    vjust = -2.2,
    size = 2.2,
    color = "black"
  ) +
  geom_text(
    aes(label = plot_label),
    vjust = 3,
    size = 1.7,
    color = "black"
  ) +
  scale_color_viridis_c(name = "Literalität") +
  scale_y_reverse(
    breaks = 0:4,
    labels = c("Gesamt", "Geschlecht", "Migration", "Deutsch", "Elternschaft"),
    limits = c(4.1, -0.1)
  ) +
  labs(
    x = NULL, y = "Gruppen",
    title = NULL
  ) +
  theme_classic() +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )

pdex

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_D_Deutsch.png"))

```



6.2.6.3 Num DE Labels

```
agg_png(here("Results", "Fig2A_D_Num_Deutsch.png"), res = 576, width = 3200, height = 2250)

df_a2num_de_X <- df_a2num_de %>%
  mutate(plot_label = case_match(plot_label,
    "Male" ~ "Männer",
    "Female" ~ "Frauen",
    "Native-born" ~ "DE geboren",
    "Foreign-born" ~ "Ausl. geboren",
    "Native" ~ "Erstsprache",
    "Non-native" ~ "nicht Erstsprache",
    "Parent" ~ "Mit",
    "Childless" ~ "Ohne")
  )

pdex <- ggplot(df_a2num_de_X, aes(x = x, y = y, color = mean)) +
  geom_segment(
    data = edges_de_num,
    aes(x = x, y = y, xend = xend, yend = yend),
    color = "grey75",
    linetype = "dotted"
  ) +
  geom_point(size = 5) +
  geom_text(
    aes(label = round(mean, 0)),
    vjust = -2.2,
    size = 2.2,
  )
```

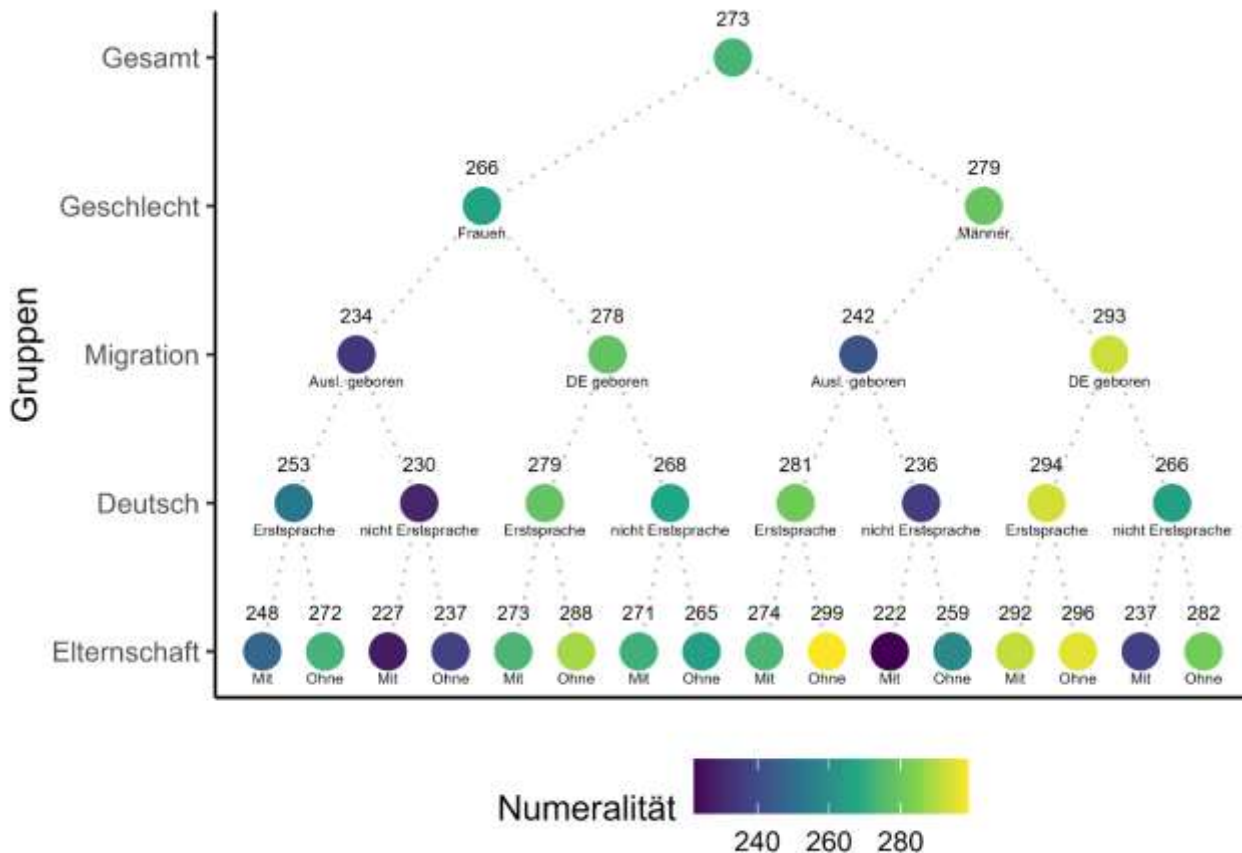
```

    color = "black"
  ) +
  geom_text(
    aes(label = plot_label),
    vjust = 3,
    size = 1.7,
    color = "black"
  ) +
  scale_color_viridis_c(name = "Numeralität") +
  scale_y_reverse(
    breaks = 0:4,
    labels = c("Gesamt", "Geschlecht", "Migration", "Deutsch", "Elternschaft"),
    limits = c(4.1,-0.1)
  ) +
  labs(
    x = NULL, y = "Gruppen",
    title = NULL
  ) +
  theme_classic() +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )

pdex

dev.off()
## png
## 2
include_graphics(here("Results", "Fig2A_D_Num_Deutsch.png"))

```



7 Tbl S2. Intersectional Descriptives (16 Groups)

The unweighted N and sec education % were moved to Fig A2 construction

```
# Fig A2 group labels
group_labels_16 <- c(
  "male_not_natlang_child", "male_not_notnatlang_child",
  "male_fb_natlang_child", "male_fb_notnatlang_child",
  "male_not_natlang_nochild", "male_not_notnatlang_nochild",
  "male_fb_natlang_nochild", "male_fb_notnatlang_nochild",
  "female_not_natlang_child", "female_not_notnatlang_child",
  "female_fb_natlang_child", "female_fb_notnatlang_child",
  "female_not_natlang_nochild", "female_not_notnatlang_nochild",
  "female_fb_natlang_nochild", "female_fb_notnatlang_nochild"
)

# Extract Literacy from Fig A2 data
df_figA2_values <- df_lit_all %>%
  filter(cycle %in% c("df2", "Cycle 2", "Cycle 2 (2022)", "2022")) %>%
  filter(group_label %in% group_labels_16) %>%
  select(group_label, Literacy_wmean = mean, ALE_pct_wmean = pct_ale)

# Calculate Unweighted N and Sec Education from df2
df2_calc <- df2 %>%
  filter(ed_currently != 1) %>%
  filter(!is.na(female) & !is.na(FORBORN) & !is.na(FORLANG) &
    !is.na(children) & !is.na(ed3)) %>%
  mutate(
```

```

    ed_sec_or_higher = ifelse(ed3 %in% c(2, 3), 1, 0),
    # labwel for joining
    group_label = paste0(
      ifelse(female == 1, "female", "male"), "_",
      ifelse(FORBORN == 1, "fb", "not"), "_",
      ifelse(FORLANG == 1, "notnatlang", "natlang"), "_",
      ifelse(children == 1, "child", "nochild")
    )
  ) %>%
  group_by(group_label, female, FORBORN, FORLANG, children) %>%
  summarise(
    N_unweighted = n(),
    SecEd_pct_wmean = weighted.mean(ed_sec_or_higher, SPFWT0, na.rm = TRUE) * 100,
    .groups = "drop"
  )

table_16_groups <- df_figA2_values %>%
  left_join(df2_calc, by = "group_label") %>%
  mutate(
    Gender = ifelse(female == 1, "Female", "Male"),
    Origin = ifelse(FORBORN == 1, "Foreign-born", "Native-born"),
    Language = ifelse(FORLANG == 1, "Non-native speaker", "Native speaker"),
    Parenthood = ifelse(children == 1, "Parent", "Childless")
  ) %>%
  select(Gender, Origin, Language, Parenthood, N_unweighted, Literacy_wmean, ALE_pct_wmean,
  SecEd_pct_wmean) %>%
  arrange(desc(Gender), desc(Origin), Language, desc(Parenthood))

table_16_out <- table_16_groups %>%
  kable("html",
    digits = 1,
    caption = "Descriptive Statistics for 16 Intersectional Groups (Cycle 2)",
    col.names = c("Gender", "Origin", "Language", "Parenthood", "N (Cases)", "Mean Lite
racy", "ALE Participation (%)", "Sec. Ed or Higher (%))" %>%
    kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))

save_kable(table_16_out, file = here::here("Results", "TS3.html"))

table_16_out

```

Descriptive Statistics for 16 Intersectional Groups (Cycle 2)

Gender	Origin	Language	Parenthood	N (Cases)	Mean Literacy Participation (%)	ALE Participation (%)	Sec. Ed or Higher (%)
Male	Native-born	Native speaker	Parent	29373	262.8	44.7	86.3
Male	Native-born	Native speaker	Childless	18353	271.8	40.4	87.2
Male	Native-born	Non-native speaker	Parent	1209	245.9	47.4	77.1
Male	Native-born	Non-native speaker	Childless	730	257.9	41.9	77.6
Male	Foreign-born	Native speaker	Parent	2074	240.5	53.4	75.9
Male	Foreign-born	Native speaker	Childless	1130	255.0	48.9	87.6
Male	Foreign-born	Non-native speaker	Parent	2595	206.9	37.4	69.3
Male	Foreign-born	Non-native speaker	Childless	1387	236.6	45.5	84.9
Female	Native-born	Native speaker	Parent	40523	263.1	43.8	87.8
Female	Native-born	Native speaker	Childless	14459	279.1	45.6	91.0
Female	Native-born	Non-native speaker	Parent	1459	242.6	39.2	74.9
Female	Native-born	Non-native speaker	Childless	652	262.8	51.9	85.4
Female	Foreign-born	Native speaker	Parent	2973	239.4	44.2	82.4
Female	Foreign-born	Native speaker	Childless	1011	259.8	46.3	87.9
Female	Foreign-born	Non-native speaker	Parent	3507	214.5	41.0	71.7
Female	Foreign-born	Non-native speaker	Childless	1244	238.9	50.0	88.9

8 Citations

```

packages %>%
  map(citation) %>%
  print(style = "text")
## [[1]]
## Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R,
## Golemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E,
## Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi
## K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the
## tidyverse." Journal of Open Source Software, 4*(43), 1686.
## doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.
##
## [[2]]
## Wickham H, Hester J, Bryan J (2024). _readr: Read Rectangular Text
## Data_. R package version 2.1.5,

```

```
## <https://CRAN.R-project.org/package=readr>.
##
## [[3]]
## Xie Y (2025). _knitr: A General-Purpose Package for Dynamic Report
## Generation in R_. R package version 1.50, <https://yihui.org/knitr/>.
##
## Xie Y (2015). _Dynamic Documents with R and knitr_, 2nd edition.
## Chapman and Hall/CRC, Boca Raton, Florida. ISBN 978-1498716963,
## <https://yihui.org/knitr/>.
##
## Xie Y (2014). "knitr: A Comprehensive Tool for Reproducible Research in
## R." In Stodden V, Leisch F, Peng RD (eds.), _Implementing Reproducible
## Computational Research_. Chapman and Hall/CRC. ISBN 978-1466561595.
##
## [[4]]
## Müller K (2020). _here: A Simpler Way to Find Your Files_. R package
## version 1.0.1, <https://CRAN.R-project.org/package=here>.
##
## [[5]]
## Wickham H (2016). _ggplot2: Elegant Graphics for Data Analysis_.
## Springer-Verlag New York. ISBN 978-3-319-24277-4,
## <https://ggplot2.tidyverse.org>.
##
## [[6]]
## Slowikowski K (2024). _ggrepel: Automatically Position Non-Overlapping
## Text Labels with 'ggplot2'_. R package version 0.9.6,
## <https://CRAN.R-project.org/package=ggrepel>.
##
## [[7]]
## Pedersen T, Shemanarev M (2024). _ragg: Graphic Devices Based on AGG_.
## R package version 1.3.3, <https://CRAN.R-project.org/package=ragg>.
##
## [[8]]
## Lumley T (2024). "survey: analysis of complex survey samples." R
## package version 4.4.
##
## Lumley T (2004). "Analysis of Complex Survey Samples." _Journal of
## Statistical Software_, *9*(1), 1-19. R package version 2.2.
##
## Lumley T (2010). _Complex Surveys: A Guide to Analysis Using R: A Guide
## to Analysis Using R_. John Wiley and Sons.
##
## [[9]]
## Zhu H (2024). _kableExtra: Construct Complex Table with 'kable' and
## Pipe Syntax_. R package version 1.4.0,
## <https://CRAN.R-project.org/package=kableExtra>.
##
## [[10]]
## Wickham H, Pedersen T, Seidel D (2023). _scales: Scale Functions for
## Visualization_. R package version 1.3.0,
## <https://CRAN.R-project.org/package=scales>.
##
## [[11]]
## Vargas Sepulveda M (2025). _wbstats: Programmatic Access to the World
## Bank API_. Oak Ridge National Laboratory, Oak Ridge, Tennessee. Vargas
## Sepulveda, Mauricio (2025). wbstats: Programmatic Access to the World
## Bank API. Oak Ridge National Laboratory. Oak Ridge, Tennessee. URL
## https://doi.org/10.11578/dc.20171025.1827,
## <https://doi.org/10.11578/dc.20171025.1827>.
```

9 Colophon

```

sessionInfo()
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] wbstats_1.1      scales_1.3.0      kableExtra_1.4.0  survey_4.4-2
## [5] survival_3.6-4   Matrix_1.7-0      ragg_1.3.3         ggrepel_0.9.6
## [9] here_1.0.1       knitr_1.50        lubridate_1.9.4    forcats_1.0.0
## [13] stringr_1.5.1    dplyr_1.1.4       purrr_1.0.4        readr_2.1.5
## [17] tidyr_1.3.1      tibble_3.2.1      ggplot2_3.5.1      tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      xfun_0.51         bslib_0.9.0       lattice_0.22-6
## [5] tzdb_0.5.0        vctrs_0.6.5       tools_4.4.1        generics_0.1.3
## [9] curl_6.2.2        parallel_4.4.1    pacman_0.5.1       pkgconfig_2.0.3
## [13] lifecycle_1.0.4   compiler_4.4.1    farver_2.1.2       textshaping_1.0.0
## [17] munsell_0.5.1     mitools_2.4       htmltools_0.5.8.1  sass_0.4.9
## [21] yaml_2.3.10       crayon_1.5.3      pillar_1.10.1      jquerylib_0.1.4
## [25] cachem_1.1.0      tidyselect_1.2.1  digest_0.6.37      stringi_1.8.4
## [29] labeling_0.4.3     splines_4.4.1     rprojroot_2.0.4    fastmap_1.2.0
## [33] colorspace_2.1-1  cli_3.6.4         magrittr_2.0.3     withr_3.0.2
## [37] bit64_4.6.0-1     timechange_0.3.0  httr_1.4.7         rmarkdown_2.29
## [41] bit_4.6.0         png_0.1-8         hms_1.1.3          evaluate_1.0.3
## [45] viridisLite_0.4.2 rlang_1.1.5       Rcpp_1.0.14        glue_1.8.0
## [49] DBI_1.2.3         xml2_1.3.8        svglite_2.1.3      rstudioapi_0.17.1
## [53] vroom_1.6.5       jsonlite_2.0.0    R6_2.6.1           systemfonts_1.2.1

```